Statistics and Its Applications

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Phase 1 Report: Association Between Sugar Intake, Smoking, and Tooth Decay

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Abstract

This study investigates the association between total sugar intake, smoking status, and tooth decay in US adults using data from the National Health and Nutrition Examination Survey (NHANES) 2021-2023. After merging demographic, dietary, oral health, and smoking datasets, a final analytical sample of 4,788 adults aged 18 and older was obtained. Descriptive analyses revealed that 37.18% of participants reported having tooth decay, and 41.33% had a history of smoking. The mean daily sugar intake was 95.47 grams (SD = 68.13). Notably, the distribution of sugar intake was highly right-skewed (skewness = 2.31), and males consumed significantly more sugar than females. These initial findings suggest potential associations that will be formally tested in subsequent phases.

1 Introduction

Tooth decay is a major public health concern, influenced by various lifestyle and dietary factors. The primary objective of this project is to answer the following research question: **Is sugar intake and smoking associated with tooth decay in US adults, even after controlling for age and gender?** This report presents the initial exploratory and descriptive findings of this investigation.

2 Methods

2.1 Data Source and Study Design

The data for this study were extracted from the National Health and Nutrition Examination Survey (NHANES), a program conducted by the Centers for Disease Control and Prevention (CDC). This is an **observational study**; therefore, causal relationships cannot be established.

2.2 Data Processing and Analytical Population

Four datasets were merged using the unique identifier `SEQN`: Demographic (`DEMO_L`), Dietary (`DR1TOT_L`), Oral Health (`OHQ_L`), and Smoking (`SMQ_L`). After filtering for adults aged 18 and older and handling missing data, the final sample included **4,788** participants.

2.3 Missing Data Audit

Before cleaning, the missing data rates were as follows: Sugar Intake (22.82%), Tooth Decay (15.78%), and Smoking (12.00%). All rows with missing values in the primary variables were excluded from the analysis.

2.4 Variables

The primary variables of interest are:

- **Response Variable:** Tooth Decay (binary: Yes/No), derived from `OHQ660`.
- **Main Explanatory Variables:** Total Sugar Intake (`DR1TSUGR`) and Smoking Status (`SMQ020`).
- **Covariates:** Age (`RIDAGEYR`) and Gender (`RIAGENDR`).
- **Survey Weight:** `WTMEC2YR` was used for weighted analyses.

3 Results

3.1 Descriptive Statistics

The analytical population consisted of 4,788 participants, including 2,110 females (55.93%) and 2,678 males (44.07%). Table 1 presents the summary statistics for sugar intake and age.

Table 1: Descriptive Statistics

Variable	Mean	Median	SD	Variance	IQR
Sugar (grams)	95.47	81.12	68.13	4641.38	72.59
Age (years)	55.05	59.00	16.84	283.53	27.00

3.2 Frequency of Tooth Decay and Smoking

- **Tooth Decay:** 1,780 participants (37.18%) reported having tooth decay, while 3,008 (62.82%) did not.
- **Smoking:** 1,979 participants (41.33%) had a history of smoking (at least 100 cigarettes), while 2,809 (58.67%) did not.

3.3 Descriptive Comparison by Gender

Table 2 shows the comparison of sugar intake between males and females. Males consumed significantly more sugar than females (107.79 g vs. 85.75 g).

Table 2: Sugar Intake by Gender

Gender	Mean (g)	Median (g)	SD
Male	107.79	91.99	78.44
Female	85.75	74.98	56.90

3.4 Weighted Analysis

The raw mean of sugar intake was 95.47 grams, while the weighted mean was 95.58 grams. The weighted proportion of smokers was approximately 34.65% of the population, and the weighted proportion of tooth decay was 34.65%.

3.5 Visualizations

Figures 1 to 14 display 14 high-quality visualizations generated for this analysis.

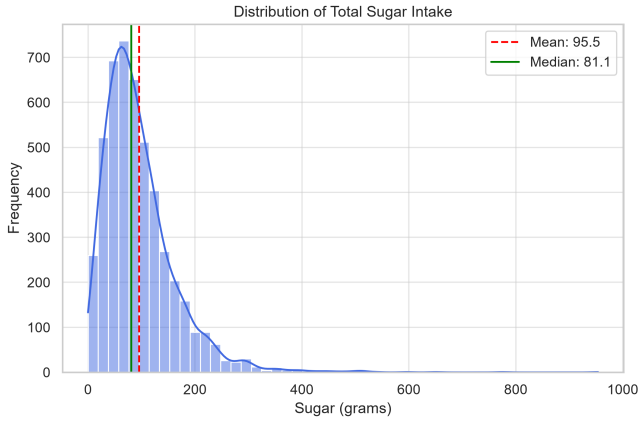


Figure 1: Distribution of Total Sugar Intake with Mean and Median Lines

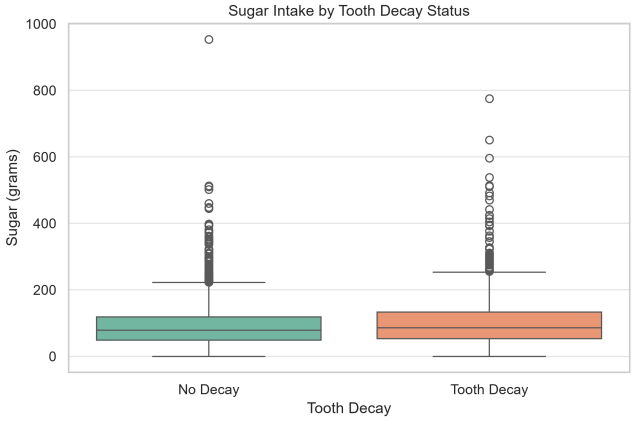


Figure 2: Sugar Intake by Tooth Decay Status

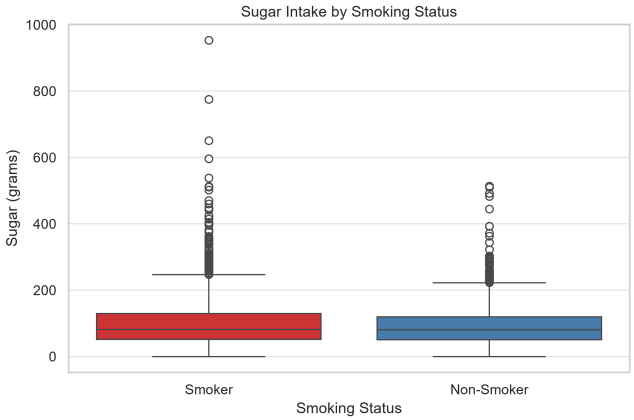


Figure 3: Sugar Intake by Smoking Status

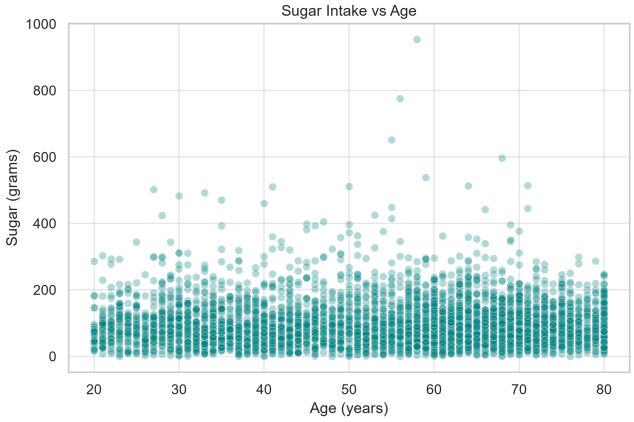


Figure 4: Sugar Intake vs Age

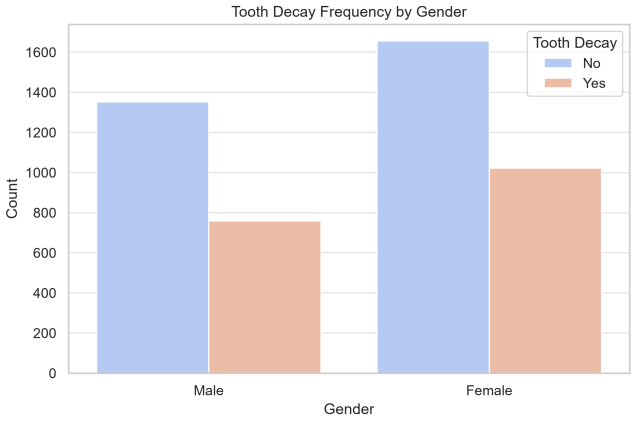


Figure 5: Tooth Decay Frequency by Gender

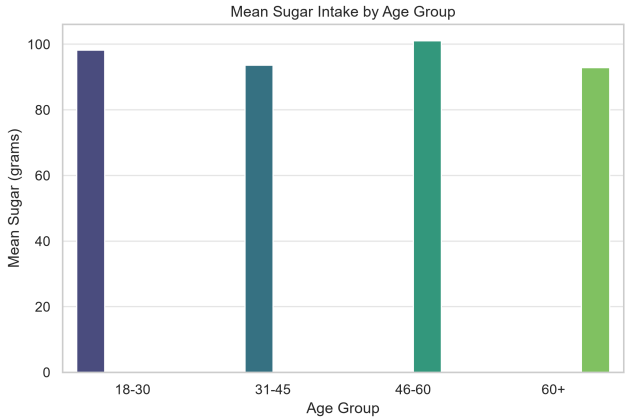


Figure 6: Mean Sugar Intake by Age Group

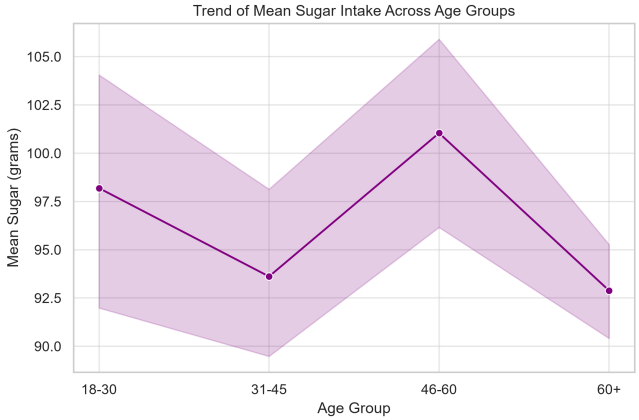


Figure 7: Trend of Mean Sugar Intake Across Age Groups

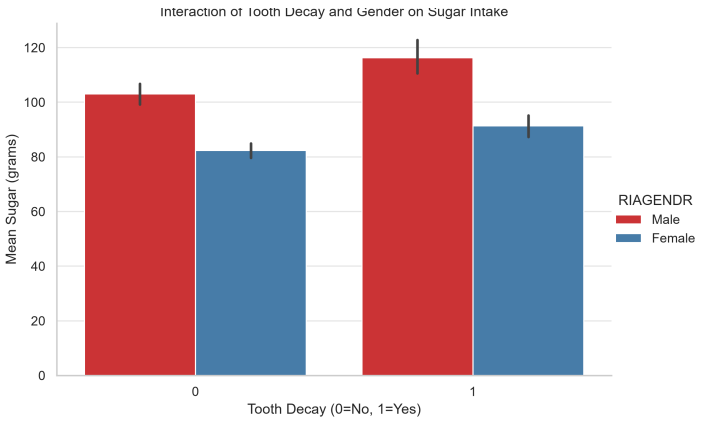


Figure 8: Interaction of Tooth Decay and Gender on Sugar Intake

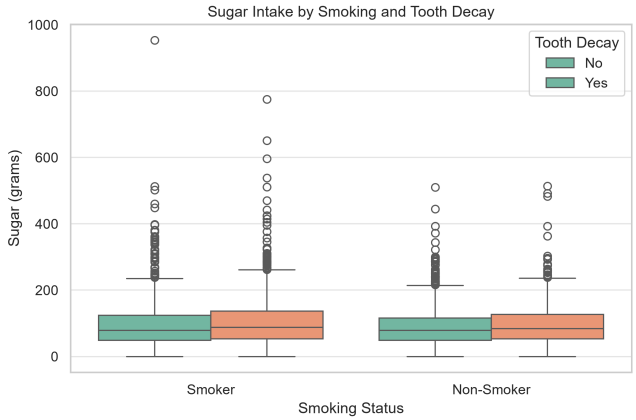


Figure 9: Sugar Intake by Smoking and Tooth Decay

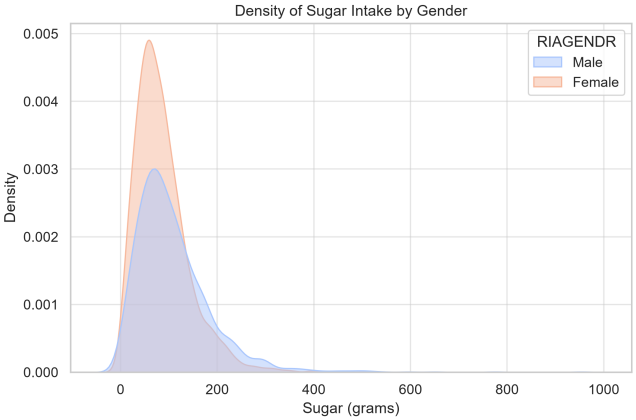


Figure 10: Density of Sugar Intake by Gender

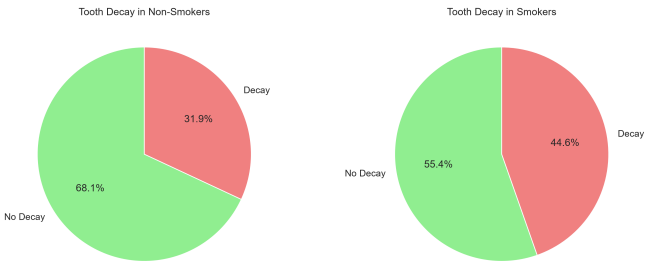


Figure 11: Tooth Decay Percentage by Smoking Status

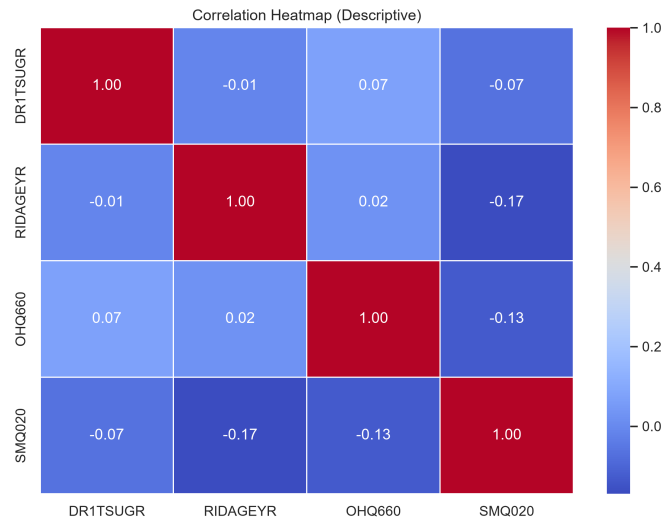


Figure 12: Correlation Heatmap (Descriptive)

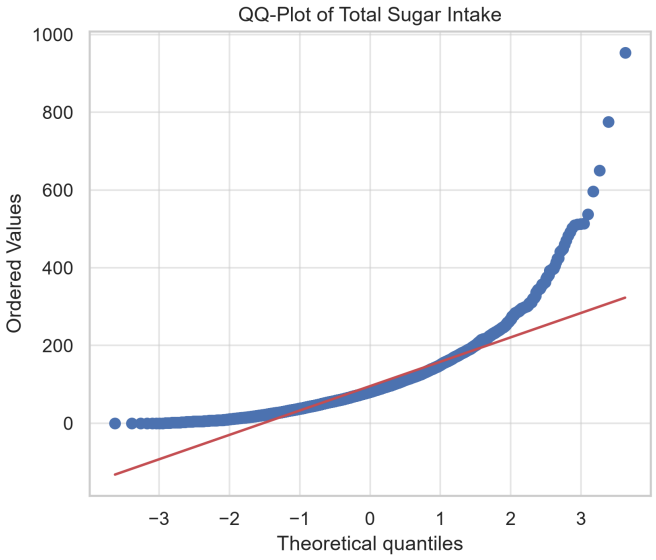


Figure 13: QQ-Plot of Total Sugar Intake

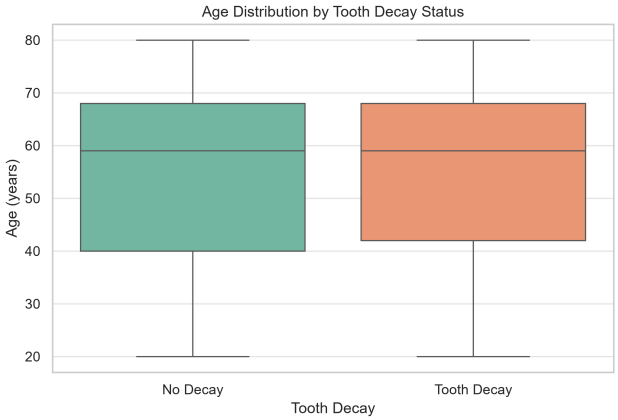


Figure 14: Age Distribution by Tooth Decay Status

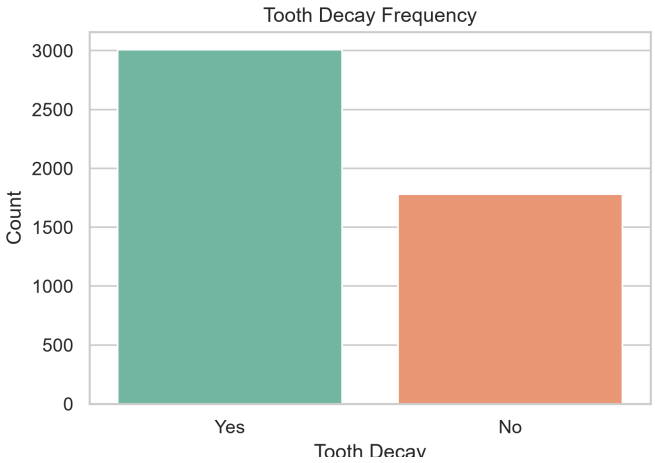


Figure 15: Frequency of Tooth Decay

4 Discussion and Conclusion

The data preparation and descriptive analysis confirmed that the merged NHANES dataset is suitable for investigating the primary research question. Initial visual and numerical evidence suggests a potential association between sugar intake, smoking, and tooth decay, with notable gender differences in sugar consumption. In the subsequent phases, formal statistical inference (hypothesis testing and regression modeling) will be conducted to rigorously test this association while controlling for covariates.